

University of South Florida

CDA 4205L Computer Architecture Lab

Lab Report

Semester: Fall 2023		
Experiment	Number:	1
	Date:	08/31/2023
Lab	Section:	001
	Lab TA:	Benjamin Kreiger
Report	Due Date:	09/07/2023
Group (Lab3 onwards)	Member #1 name:	Diya Jain
	Member #2 name:	

ABSTRACT:

The purpose of this lab was to get introduced to RISC-V and RARS. This lab also included the practical implementation of RISC-V.

INTRODUCTION:

In this lab, I had to print different statements and learn how to take input from the user and input that in the console. Firstly, we had to download RARS on our computer. We already had a prewritten code that would help us go through all the required steps.

METHOD:

Software Tools Used:

1. RARS (Runtime Simulator)
2. RISC-V

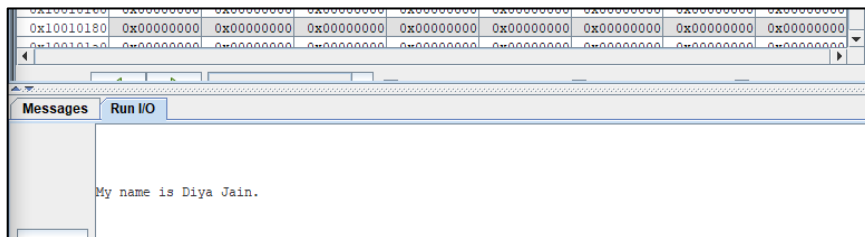
Files:

There are no files, but I have added a picture of my code.

RESULTS:

T1: Modify the code so that it prints your name instead of “Hello World!”.

This is a picture that shows that my name is printed in the console.



T2: What do ecalls 4 and 10 do? What is the purpose of registers a7 and a0?

ecall 4: Prints a null-terminated string to the console.

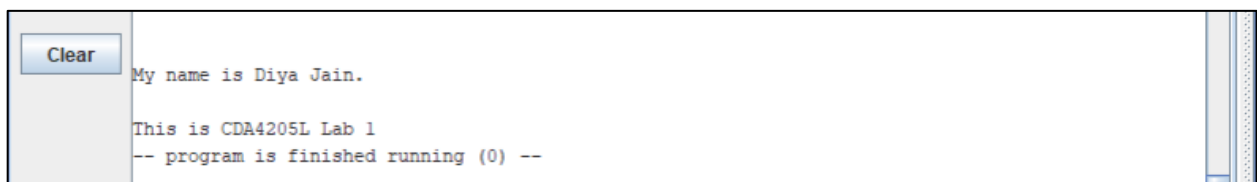
ecall 10: Exits the program with code 0.

The purpose of register a7 is to store the value of ecall 4 in it which defines the 'print string' ecall

The purpose of register a0 is to store the address of label "msg" in the argument register a0.

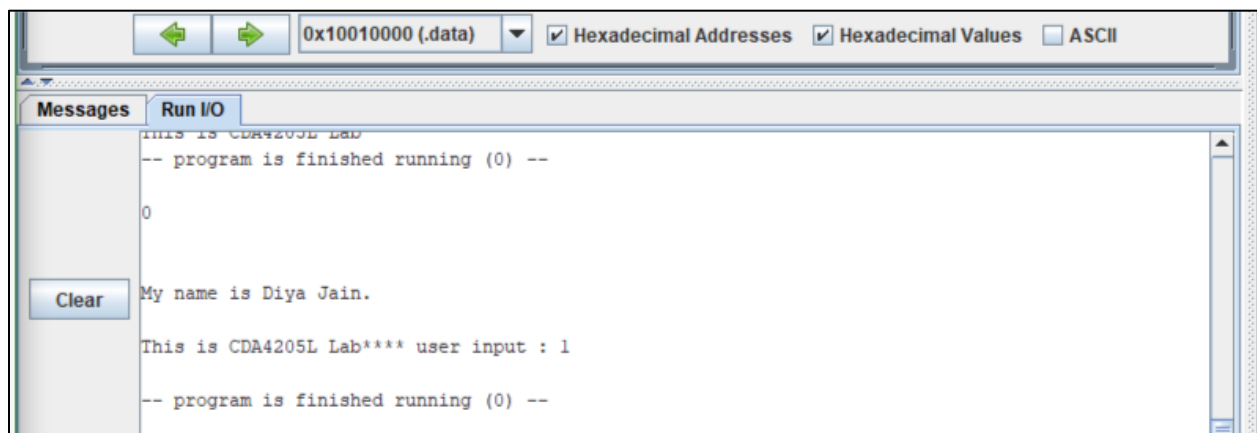
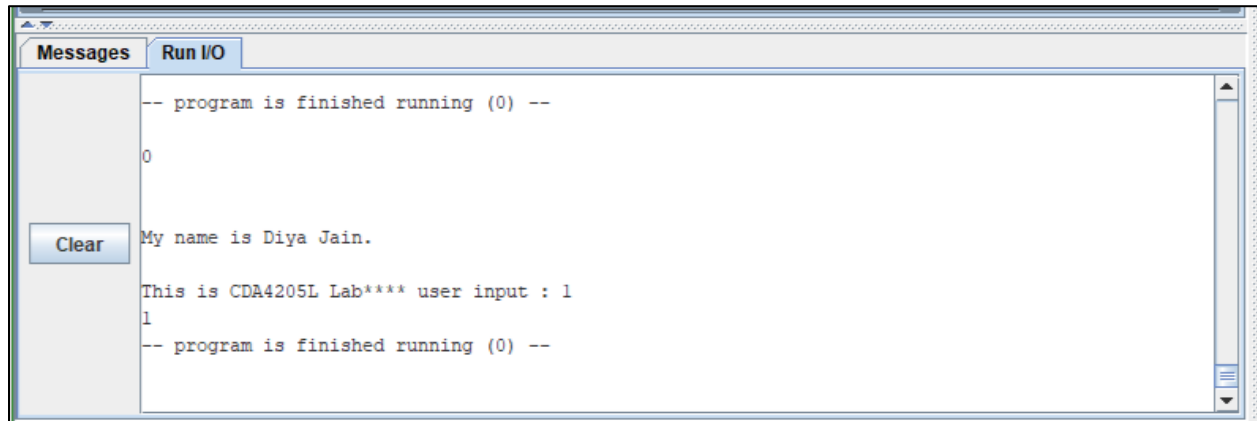
T3: Modify the code so that, in addition to your name, it prints “This is CDA 4205L Lab 1”

This picture shows the message being printed in the console.



T4: Modify the code so that the user can input the number at runtime.

The below pictures show two ways in which the user input was showing in the console.



T5: Snippet of code.

```
helloRISCV.asm*  
1  # "Hello World" program in RISC-V assembly  
2  # Information on ecalls: https://github.com/TheThirdOne/ra  
3  
4  .globl __main  
5  
6  .data  
7      msg: .string "\n\nMy name is Diya Jain.\n"  
8      msg2: .string "\nThis is CDA4205L Lab"  
9      msg3: .string ""  
10  
11 .text  
12  
13     __main:  
14  
15         li a7, 1  
16         addi a1, a7, 0  
17         ecall  
18  
19         li a7, 4  
20         la a0, msg  
21         ecall
```

```
20  
21         li a7, 4  
22         la a0, msg2  
23         ecall  
24  
25         li a7, 5  
26         ecall  
27  
28         li a7, 1  
29         ecall  
30  
31  
32         li a7, 10  
33         ecall  
34
```

DISCUSSION:

The implications from doing this lab were that we got a refresher on RISC-V and how to write code and actually implement it. There was one problem while running the program which was that whenever we used the pop-up dialog for the user input, it would mention user input in the output console. It would have been better without the 'user input' showing in the console.

CONCLUSION:

This lab was an introduction to RISC-V and RARS. We got to print statements and learned how to take user input using a different ecall. We also learned about the concept of ecall and what they mean when you use a certain number.